Department of Mathematics, IIT Madras

MA 2102 - Differential Equations, Odd Semester 2026-27

Assignment-2 : Second Order ODEs

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1. Show that $y = c_1e^{-x} + c_2e^{2x}$ is the general solution of $y'' - y' - 2y = 0$ on any interval.

2. Verify that $y = c_1x^{-1} + c_2x^5$ is the solution of $x^2y'' - 3xy' - 5y = 0$ on any interval $[a, b]$ that does not contain the origin.

3. Consider the differential equation $x^2y'' + xy' - 4y = 0$.
   (a) Show that x^2 and $1/x^2$ are linearly independent solutions of it on the interval $0 < x < \infty$.
   (b) Write the general solution of the given equation.
   (c) Find the solution that satisfies the conditions $y(2) = 3$, $y'(2) = -1$. Explain why this solution is unique. Over what interval is this solution defined?

4. Given that x, x^2 and x^4 are all solutions of $x^3y''' - 4x^2y'' + 8xy' - 8y = 0$. Also show that they are linearly independent on $0 < x < \infty$ and write the general solution.

5. Define the Wronskian $W(y_1, y_2)$ of any two differentiable functions y_1 and y_2 defined in an interval $(a, b) \subset \mathbb{R}$. Show that $W(y_1, y_2) = 0$ if y_1 and y_2 are linearly dependent.

6. If y_1 and y_2 are any two solutions of a second order linear homogeneous ordinary differential equation which is defined in an interval $(a, b) \subset \mathbb{R}$, then $W(y_1, y_2)$ is either identically zero or non-zero at any point of the interval (a, b).

7. Consider the two functions $f(x) = x^3$ and $g(x) = x^2|x|$ on the interval $[-1, 1]$.
   (a) Show that their Wronskian $W(f, g)$ vanishes identically.
   (b) Show that f and g are not linearly dependent.

8. Use inspection or experiment to find a particular solution of the following equations:
   $(a)\ y'' - 2y = \sin x; \quad (b)\ y'' - 2y' = 6.$

9. Find the general solution of the following equations where one solution y_1 is given:
   $(a)\ x^2y'' + xy' - y = 0,\ y_1 = x; \qquad (b)\ y'' + y = 0,\ y_1 = \sin x;$
   $(c)\ (1 - x)^2y'' - 2xy' + 2y = 0,\ y_1 = x; \quad (d)\ x^2y'' + xy' + (x^2 - \frac{1}{4})y = 0,\ y_1 = x^{(-1/2)} \sin x;$
   $(e)\ xy'' - (2x + 1)y' + (x + 1)y = 0,\ y_1 = e^x; \quad (f)\ (x - 1)y'' - xy' + y = 0,\ y_1 = x.$

10. Solve the following:
    $(a)\ y'' + 3y' - 54y = 0; \quad (b)\ y'' + 4y' + 4y = 0;$
    $(c)\ y'' - 6y' + 13y = 0; \quad (d)\ y'' - 9y' + 20y = 0;$

11. Solve the following IVPs.

$(a)\ 4y'' + 24y' + 37y = 0, y(\pi) = 1, y'(\pi) = 0.$

$(b)\ 16y'' - 40y' + 25y = 0,\ y(0) = 3, y'(0) = -9/4.$

$(c)\ y'' - 5y' + 6y = 0,\ y(1) = e^2, y'(1) = 3e^2;$

$(d)\ y'' + 8y' - 9y = 0,\ y(1) = 2, y'(1) = 0;$

$(e)\ y'' + 4y' + 5y = 0,\ y(0) = 1, y'(0) = 0.$

12. Solve the following problems by using the method of undetermined coefficients:

$(a)\ y'' - 3y' + 2y = 14\sin 2x - 18\cos 2x;$ $(b)\ y'' - 4y' + y = x^2 - 2x + 2;$

$(c)\ y'' + 9y = 2\sin 3x + 4\sin x - 26e^{-2x} + 27x^3;$ $(d)\ y'' - 2y' + 2y = e^x \sin x;$

$(e)\ y'' - 2y' + y = xe^x;$ $(f)\ y'' + 10y' + 25y = 14e^{-5x};$

$(g)\ y'' - 3y' = x + e^x \sin x$ $(h)\ y'' + 2y = e^x + 2.$

13. Solve the following problems by using the method of variation of parameters:

$(a)\ y'' + 2y' + y = \frac{e^{-x}}{x^2};$ $(b)\ y'' + a^2 y = \sec ax;$ $(c)\ y'' - 3y' + 2y = \frac{e^x}{1+e^x};$

$(d)\ y'' - 2y' - 3y = 64xe^{-x};$ $(e)\ y'' + 2y' + y = e^{-x}\log x;$ $(f)\ 2y'' + 3y' + y = ^{-3x};$

$(g)\ y'' - 4y' + 3y = (1 + e^{-x})^{-1};$ $(h)\ y'' + y = \sec x \tan x;$ $(i)\ x^2y'' - 4xy' + 6y = 21x^{-4};$

$(j)\ x^2y'' - 2xy' + 2y = x^3 \cos x;$ $(k)\quad xy'' - y' = (3 + x)x^2e^x;$

$(l)\ y'' + \frac{1}{x}y' - \frac{1}{x^2}y = \log x$, given that $y_1 = x$ and $y_2 = 1/x$ are two linearly independent solutions of its reduced equation;

$(m)\ (x^2 + x)y'' + (2 - x^2)y' - (2 + x)y = x(x + 1)^2$, if the complementary function is $c_1e^x + c_2/x$.

14. Find the general solution of each of the following equations: (Note $D^n \equiv \frac{d^n}{dx^n}$).

(a) $(D^3 - 4D^2 + 5D - 2)y = 0;$

(b) $(D^2 - 3D + 2)y = (4x + 5)e^{3x};$

(c) $(D^2 - 2D + 3)y = 3e^{-x}\cos x;$

(d) $(D^3 - 2D^2 - 5D + 6)y = 18e^x;$

(e) $(D^2 + 25)y = 50\cos 5x + 30\sin 5x;$

(f) $(D^3 - 2D^2 + 4D - 8)y = 8(x^2 + \cos 2x);$

(g) $(D^3 + 3D^2 - 4)y = 12e^{-2x} + 9e^x.$